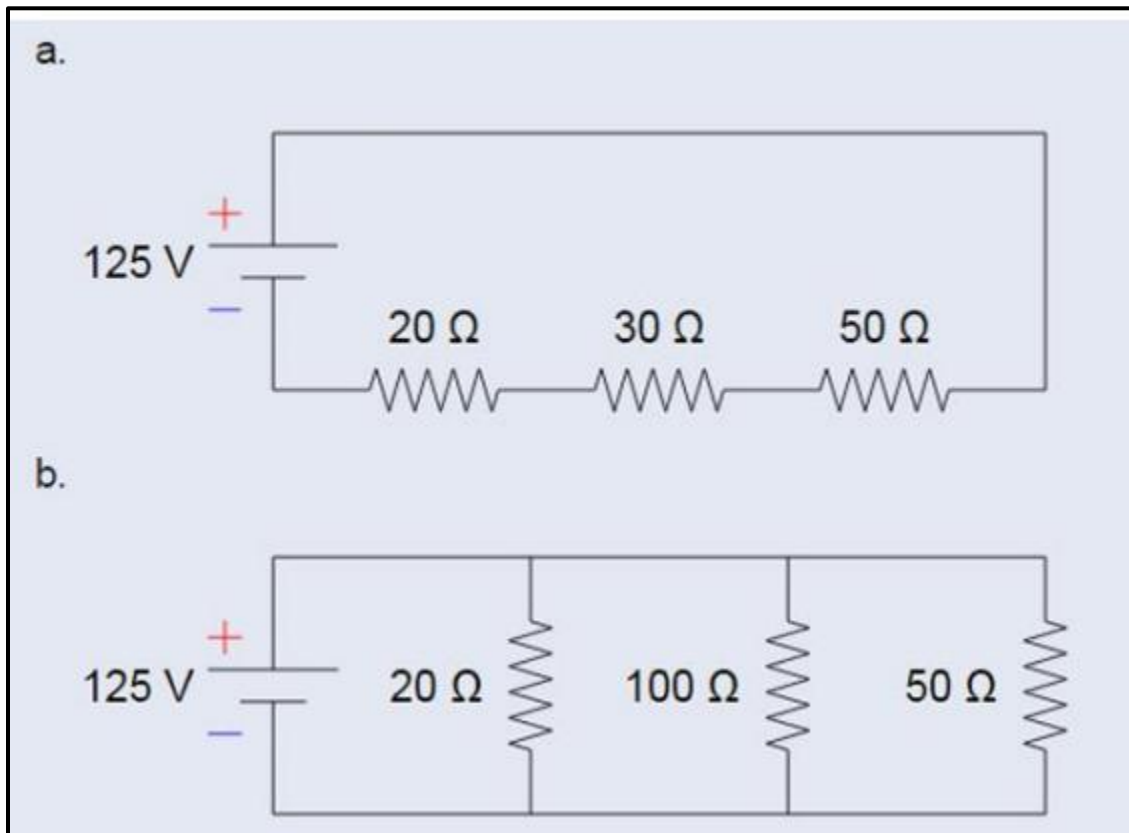


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ICS 4.1 B April – July 2026

ICS 4111: Embedded Systems & IoT - Practical Exercise 2

Part A: Circuit Calculations



Circuit A: Series Configuration

1. Calculate the total resistance in ohms

In a series circuit, total resistance (R_t) is the sum of all individual resistances:

$$R_t = R_1 + R_2 + R_3$$

$$R_t = 20\Omega + 30\Omega + 50\Omega = 100\Omega$$

2. Calculate the total current from the power supply

Using Ohm's Law ($I = V / R$):

$$I_{\text{total}} = 125\text{V} / 100\Omega = 1.25\text{A}$$

3. Calculate the current through each resistor

In a series circuit, the current remains constant at every point:

$$I_{\text{R1}} = 1.25\text{A}$$

$$I_{\text{R2}} = 1.25\text{A}$$

$$I_{\text{R3}} = 1.25\text{A}$$

4. Calculate the voltage drop through each resistor

Using $V = I \times R$:

$$V_{\text{R1}} = 1.25\text{A} \times 20\Omega = 25\text{V}$$

$$V_{\text{R2}} = 1.25\text{A} \times 30\Omega = 37.5\text{V}$$

$$V_{\text{R3}} = 1.25\text{A} \times 50\Omega = 62.5\text{V}$$

$$\text{Verification: } 25\text{V} + 37.5\text{V} + 62.5\text{V} = 125\text{V}$$

5. Calculate the power dissipated through each resistor

Power = Voltage $\times$ Current

$$P_{\text{R1}} = 25\text{V} \times 1.25\text{A} = 31.25\text{W}$$

$$P_{\text{R2}} = 37.5\text{V} \times 1.25\text{A} = 46.875\text{W}$$

$$P_{\text{R3}} = 62.5\text{V} \times 1.25\text{A} = 78.125\text{W}$$

Circuit B: Parallel Configuration

1. Calculate the total resistance in ohms

Using the reciprocal formula for parallel resistance:

$$1/R_t = 1/R_4 + 1/R_5 + 1/R_6$$

$$1/R_t = 1/20 + 1/100 + 1/50 = (5 + 1 + 2)/100 = 8/100$$

$$R_t = 100 / 8 = 12.5\Omega$$

2. Calculate the total current from the power supply

Using Ohm's Law:

$$I_{\text{total}} = 125\text{V} / 12.5\Omega = 10\text{A}$$

3. Calculate the current through each resistor

Using $I = V / R$ for each branch:

$$I_{\text{R4}} = 125\text{V} / 20\Omega = 6.25\text{A}$$

$$I_{\text{R5}} = 125\text{V} / 100\Omega = 1.25\text{A}$$

$$I_{\text{R6}} = 125\text{V} / 50\Omega = 2.5\text{A}$$

$$\text{Verification: } 6.25\text{A} + 1.25\text{A} + 2.5\text{A} = 10\text{A}$$

4. Calculate the voltage drop through each resistor

In a parallel circuit, the voltage across each branch is equal to the source voltage:

$$V_{\text{R4}} = V_{\text{R5}} = V_{\text{R6}} = 125\text{V}$$

5. Calculate the power dissipated through each resistor

Power = Voltage $\times$ Current

$$P_{\text{R4}} = 125\text{V} \times 6.25\text{A} = 781.25\text{W}$$

$$P_{\text{R5}} = 125\text{V} \times 1.25\text{A} = 156.25\text{W}$$

$$P_{\text{R6}} = 125\text{V} \times 2.5\text{A} = 312.5\text{W}$$